

SEFire Lab 1 Outline

Team Sapphire

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1. Introduction

Managing financial responsibilities in shared households can be difficult. Tracking who pays for rent, utilities, groceries, and other shared expenses can be confusing and may lead to missed payments or disagreements.

1.1. Societal problem FINRA Reference

In a 2024 survey, 44% of the respondents reported that it was not at all difficult to cover expenses or pay bills (Lin et al., 2025). In contrast, this implies that 56% of respondents found it difficult to pay their bills.

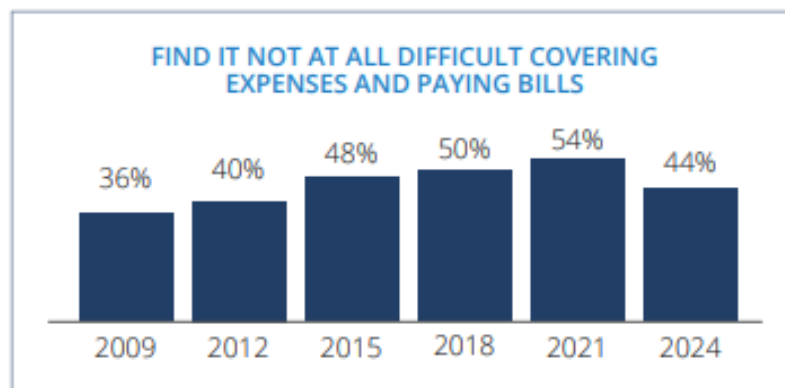


Figure 1: The difficulty in covering expenses and paying bills

1.2. Societal Problem SpareRoom Reference

Money can also cause conflict between roommates. Of 1,257 U.S. roommates surveyed by SpareRoom, 26% of disputes were related to money or utility bills (SpareRoom, 2025).

1.3. Needed Solution Characteristics:

1.3.1. What should it do?

The solution should provide one place for household members to record shared expenses, identify who paid, and track how much each person owes.

1.3.2. How should it do it?

It should support equal and custom expense splitting, automatically calculate each member's share, and display updated balances.

Users should also be able to record completed payments and receive reminders about upcoming bills and unpaid expenses.

1.3.5. Who does it help?

It should help users understand changing household costs through expected versus actual expense comparisons, budgeting guidance, financial vocabulary, and saving goals.

1.3.6. Where would this be used?

The solution should be easy to use across supported devices and restrict access to household information to authorized users.

1.4. Proposed Product: SEFire**1.4.1. SEFire in detail**

SEFire is a shared expense tracking application designed for roommates, couples, and families who manage household costs together.

1.4.2. What can users do?

Users will be able to create household groups, enter expenses, customize how costs are divided, view outstanding balances, and record completed payments.

1.4.3. SEFire's additional features

SEFire will combine expense tracking with financial vocabulary tooltips, budgeting guidance, and saving goals to support users' understanding of their shared finances.

1.4.4. What is SEFire's Goal?

The application aims to reduce confusion and disagreements by maintaining a shared record of expenses and payment status. Payments will occur outside of SEFire, and users will record their completion in the application.

1.5. Prototype Demonstration:**1.5.1. What information will the prototype use?**

The planned prototype will use a sample household and example expenses to demonstrate SEFire's main features.

1.5.2. What will the prototype do?

A sample scenario will show a user entering a household bill, selecting the participating members, and choosing an equal or custom split. The resulting balances will demonstrate how SEFire calculates each member's responsibility.

1.5.3. What will the prototype demonstrate?

The demonstration will show how payment records and outstanding balances change after users record payments completed outside the application.

1.5.4. Additional Demonstrations

Additional planned demonstrations include uploading a receipt, reviewing and correcting OCR results, and displaying reminders for upcoming or unpaid expenses. Financial vocabulary tooltips and a sample savings goal will also be added to demonstrate how SEFire supports financial awareness along with shared expense tracking.

2. SEFire Product Description

2.1. Product Overview

SEFire is a web application that gives roommates, families, and couples an easier way to manage shared expenses together.

2.2. Replacing Outdated Methods of Expense Tracking

Instead of relying on memory, text messages, and verbal communication, users can log expenses, split payments between household members, and check who still has payments due.

2.3. Reducing the Manual and Mental Effort

Receipt uploads and payment tracking make financial management less manual and mentally taxing.

2.4. Providing Financial Literacy Support

SEFire provides financial and budgeting vocabulary definitions to improve the users' financial literacy.

2.5. Product Goals

SEFire aims to reduce the confusion, stress, and manual overhead that shared expenses often create, and give users tools to better understand and manage their finances.

2.6. Key Product Features and Capabilities

2.6.1 Household groups

Users can create a household group and invite members to manage shared expenses together.

2.6.2 Expense Tracking

Users can record bills and purchases, including the amount paid and which members share the cost.

2.6.3 Custom Expense Splitting

Expenses can be divided equally or using custom amounts or percentages. SEFire calculates each member's share to reduce manual calculation.

2.6.4 Variable Expenses

Users can compare expected and actual costs and update recurring expenses when amounts change, helping members prepare for changing household bills.

2.6.5 Receipt Uploads and OCR

Users can upload receipts and extract expense details through OCR. They can review and correct the information before saving it.

2.6.6 Balances and Payment Records

Members can view outstanding balances and record payments completed outside the application, providing a shared record of who still owes money.

2.6.7 Payment Reminders

Reminders notify members about upcoming bills, subscriptions, and unpaid shares to help them keep track of their responsibilities.

2.6.8 Financial Vocabulary and Budgeting Guidance

Tooltips explain unfamiliar financial terms while budgeting guidance helps users understand and plan for shared expenses.

2.6.9 Saving Goals

Users can establish savings goals such as saving for a shared trip and track their recorded progress towards those goals.

2.7. Major Components (Hardware/Software)

2.7.1 *Hardware:*

2.7.1.1 User Devices:

The application will primarily use existing consumer hardware, i.e a smartphone, mobile device, or tablet.

2.7.1.2 Hosting Infrastructure:

Cloud servers and storage supporting Vercel and Supabase will host the web application, backend services, database, and receipt files.

2.7.2 *Software:*

2.7.2.1 Web/App Interface:

React Native will be used to develop the application's interface, display it in web browsers across different devices, and allow users to interact with it.

2.7.2.2 Backend/Application Server:

Processes requests, performs calculations, and communicates with Supabase and the PostgreSQL database.

2.7.2.3 Database and Storage:

PostgreSQL will store household membership, expenses, splits, payment records, and savings goals. Supabase Storage will store receipt images linked to expense records.

2.7.2.4 OCR Processing:

Tesseract.js. Is proposed to run Tesseract OCR in the web browser. It will extract text into suggested expense fields for users to review and correct before saving.

2.7.2.5 Authentication and Security:

Protects user accounts from unauthorized access and ensures that only users can access their own financial information.

2.7.2.6 Financial Management Features:

Provide transaction tracking, budgeting, spending analysis, and financial summaries.

Together, these components form the overall architecture of the finance application SEFire.

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3. Identification of Case Study

3.1 WHO:

Robert, a 21-year-old who is rooming with two friends, Jerry, a 22-year-old, and Bob, a 23-year-old, in an apartment. They split expenses such as rent, utilities, subscriptions, and groceries.

3.2 PROBLEM:

Robert and his roommates need a clear way to track who paid each bill, how much each person owes, and which payments have been completed.

3.3 FEATURES:

Entering expenses, custom splitting, and marking completed payments.

3.4 SCENARIO:

Robert pays a \$100 electricity bill for the apartment and enters the expense into SEFire, which is then divided into percentages based on custom splitting. They agreed on a 44/34/22 split. SEFire assigns \$44 to Robert, who made the initial purchase, and marks it paid. SEFire shows Jerry owes \$34 and Bob \$22. Upon payment, SEFire marks them as paid, and the request is archived.

3.5 FUTURE:

SEFire would be spread by word of mouth to friends and neighbors who regularly divide expenses.

4. Glossary

API — Application Programming Interface

Git — Distributed version control system

GitHub — Software development and repository-hosting platform

MFCD — Major Functional Component Diagram

OCR — Optical Character Recognition

PostgreSQL — Relational database management system

React Native — Framework for developing application interfaces

React Native for Web — Technology for using React Native components in web browsers

SEFire — Shared household Expense management application

Supabase — Backend development platform

Tesseract OCR — Optical character recognition engine

UI — User interface, which is the point of contact and interaction between a human and a device, program, or website.

Vercel — Web application hosting and deployment platform

VS Code — Visual Studio Code

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